

CO-CHALLENGED IN AN ENVIRONMENT

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This paper discusses issues related to providing challenge by drawing upon teaching experiences at a compulsory school in India. It argues for an environment in which the student, the teacher and the school are co-challenged. It deliberates on the role of the teacher and the social environment in a school within which it is possible to provide challenge in mathematics.

The challenge

In addressing the issue of providing challenge I speak from my experience in the middle and high grades. I speak in first person for two reasons: firstly I offer an interpretation which is mine; secondly the profile of a teacher in India needs to be understood as many roles rolled up in one: pedagogue, teacher, administrator and innovator to name a few. These demands are made upon a person whose primary authority stems from being a subject expert, where special attention to mathematics education and its socialisation are virtually non-existent. In the absence of a systemic backing the demand of addressing such issues which none the less exist is at a personal level. The premise of course is that the school cares and makes the providing of challenge in mathematics possible *in addition*.

I discuss how such an eventuality was possible in my school that challenged the fossilised routines of schooling and made challenge in mathematics the last mile run; inseparable in a practice in which the student, the teacher and the school were co-challenged. Adopting such an embodied perspective, I argue that it is not possible to provide challenge unless there is implicit within praxis the understanding and value of recognising, accepting and addressing challenges.

As different from the nature of challenge that a mathematical question can provide it is my interest to discuss the ways in which a teacher can call to question, dispute or challenge the current knowledge and abilities of a student. Taking a Vygotskian perspective I address the role of a teacher as embedded in an environment which possesses the ideal of questioning and challenge. As a consequence I explore some relationships that existed between participants and practices that enabled such a school culture to be constituted.

I first describe the social environment of my school. I follow this by examples from my teaching practice giving both the context and nature of challenge in mathematics that I was able to provide. I reflect on the importance of mentoring in such a school and the related nature of learning that was implicitly possible to promote. I extend my arguments in a coda, with related experiences had while conducting an empirical classroom study at a school in Norway.

Our school

Our school was run by its founder, inspired by the Indian philosopher and thinker Jiddu Krishnamurti. An experimental school, it drew on her abilities as a gifted teacher and able teacher trainer. A compulsory school it prepared students for a recognised school leaving examination at grade 10. Privately run and not funded, we may have paid a price for not wanting to be dictated externally about our goals and means. A teacher's school, these very goals and means were under constant review and debate in practice.

I was drawn by the opportunity to experiment and bring about change in a system that appeared 'different', as we put in practice equal enrolment of boys and girls, no formal examinations till the 8th grade, no raking of students, no student prefects and no school uniform to adhere to. Our morning assemblies were not stereotyped; interesting topics were shared and brought to discussion by teachers. In such a practice we questioned the norm.

Apart from visible differences there were deeper issues of not making the students compete but cooperate. As teachers we had the freedom to not follow 'the' textbook and bring in other material. We attempted to make learning rewarding and tried to do away with awards and punishment. We encouraged every student to question, be responsible and speak for herself or himself. In addressing some of the many irritants that came in the way of schooling we were left with a focus on learning and teaching alone, which was not only creative and difficult but also enjoyable and frustrating.

Our practice was far from smooth. There were issues that came up all the time which had to be addressed and addressed together: teacher and teacher, student and student, teacher and student. The underlying philosophy was one intended to recognise that to live was to be related (Krishnamurti, 1974), bring order within oneself and be responsible for one's own transformation (As in Shotton, 1998). One was to allow experience to awaken intelligence, free ourselves from fear by understanding ourselves and see the individual as important and not the system. Education was to inquire, not cease to question, see the significance of life as a whole and integrate this in the understanding of oneself (Krishnamurti, 1953).

It goes without saying that our school itself was immersed in other sources of values in our culture. As example I include folk tales: which conveyed how the tiny could overcome the mighty, epic: in which it was expected of an archer to see in his aim only the target and nothing else, and philosophy: which distinguished between knowledge as of things, acts and relations and wisdom as that of the creative self beyond these. In general participation of a person leading to knowledge revolved around the qualities of the mind in search for knowledge, and not knowledge per se. The simple metaphor was one of moving from darkness to light. The ambient envelope was one of non-violence.

At morning assemblies

Envisioning school in the way outlined, our practices accommodated debate and welcomed challenge. I begin with the ‘morning assembly’ our earliest meeting in school. Students from grade 3 onwards would assemble in two or three separate gatherings (grades 3-5, 6-7, 8-10 or 6-10). Being a small school of about 650 students, about 150 to 250 students and their teachers gathered together. Lasting 20 minutes we started the day with a skit, discussion on any topic of relevance or a mathematical problem. I elaborate on the last option and delve on how the problems I shared developed with my experience.

To appeal to a wider than classroom audience, my problems addressed students of mixed attitudes and abilities. In either gathering I would initially present for discussion an anecdote or topic that could be resolved in 15-20 minutes e.g. Archimedes’ revenge, palindromic numbers. Deliberation at these assemblies allowed students and teachers to consider views that were more or less advanced or divergent from their own. Presented outside the framework of the classroom such an effort enabled for the dissolving of barriers both within and across the grades in which we were all separated. I was able to steer discussion with those who would be my students in the future and those who were in the past.

With confidence resulting from experience, I brought to focus issues that related to the learning difficulties of students in my mathematics class. On one occasion we discussed what we all knew about the Eratosthenes’ sieve. At the middle school level this was a simple challenge, yet had benefits in my being able to extend the underlying concept with students having difficulty in finding the LCM of algebraic denominators. From a grid of many algebraic terms, such a model led us to find that term which was say a multiple of both $3x^2$ and $4xy$.

Setting more open ended challenges came with my understanding of what the practice of morning assembly could be made to *mean* within school praxis. For example I discussed the problem of Hailstone numbers (Gade, 2004):

Step 1: Take any number to start with.

Step 2: If the number is odd, triple it and add one; and if the number is even half it.

Step 3: Repeat Step 2 with the number which is resulted from performing Step 2.

Such a problem had many students suggesting numbers that I would work with at the blackboard. After a few attempts, numbers that surfaced in Step 2 were no longer proposed as the starting number in Step 1. Apart from setting the day to a fun and refreshing start I would conclude say as follows:

‘I hope you enjoyed this problem ... would you try the number 27 for me later?’

The challenge I had set would find response in students approaching me later in the day, with either solutions or excitement about related challenges they had set for each other. I had challenges in reserve for these students to attempt.

Within the classroom

The excitement of the morning assembly problem (as it got called) would sometimes spill into the class I taught and at other times yield no visible student attempts. Yet our wanting student response as teachers, brought into my thinking the need of making a problem inviting in addition to its being challenging. The classroom provided greater control and proximity for this that I learnt to utilise. I elaborate two such ways that I employed with examples.

The first example relates to challenging the practice of pencil and paper tests in evaluating students. During the teaching-learning of surface area and volume of regular solids, we first discussed various parameters of cardboard cartons of toothpaste or soap brought by the students to the class. The students then learnt to draw and make such solids from a net of faces. For the ensuing school test I asked for the construction of a solid with twice the dimensions of the carton that was brought. Student evaluation was based on their making the solid and an informal interview on related processes and reasoning. Such questioning belied the routine examination and made the related learning inviting.

My second example relates to a challenge offered within classroom teaching-learning. I refer in particular to the stage within the topic of Simple Interest when the use of the formula ($\text{Simple Interest} = \text{PTR}/100$) had become routinised for the students. Or had it? On my asking, the students and I wrote on four different strips of paper a value each for Principal, Time, Rate and Interest. We folded these strips and put them in a bag which we then shuffled. From the bag we picked a combination of three unrelated values and found the fourth and missing quantity. I offer some of the values the students gave:

SI = Rs 5000	R = $99\frac{1}{2}\%$	T = 150 years
P = Rs 4,444	SI = Rs 50,00,000	I = Rs 1,400
R = 86 %	P = Rs 1,03,109	R = $33\frac{1}{3}\%$
T = $\frac{1}{4}$ day	T = 3 hours	P = Rs 1
R = 2%	R = 150 %	SI = Rs 4,444

Apart from coming up with enough problems to work in class and beat the routine of the solving questions from the textbook, the above practice enabled students to pose questions as problems. I must add that the above values need to be viewed within a context and practice of calculations with paper and pencil alone, with the use of the calculator not incorporated in classroom teaching-learning. Thus while offering values that were to form questions; students were aware of the nature of difficulties involved in their calculation. From such a perspective the values offered reflect multiple intents; some were easy to perform calculations with while others were known to create difficulty. In addition to being inviting and challenging such a practice enabled a sense of ownership and made students active participants in teaching-learning.

While teaching

In relating examples that brought about challenge in classroom teaching-learning I had two intentions. Besides pointing to the nature of challenge that could be provided in the classroom, my intention was to draw attention to the role of a teacher since I was not the only teacher trying to innovate. Other teachers accepted such a challenge in the subjects they were teaching as well.

As a teacher-trainer our founder made inquiry and not ceasing to question part of school praxis. She led an open school and invited discussion by anyone: teacher, student or parent. She welcomed new ideas and sought them in her own as well as in the teaching of us teachers. She took part in the debates at morning assemblies and insisted learning be fun for the students. The re-interpretation of these guidelines in the subjects we taught endeared our creativity. Though this was demanding on some teachers and in certain topics, such an impetus did invite us to experiment. The freedom to be innovative was in turn given in trust. By this I mean that our need to experiment and innovate was made welcome by accepting that success and failure were a part of such attempts.

I found discovery a worthy motivation for students. Setting appropriate tasks included e.g. folding circles cut out of paper to obtain a measure of 60 degrees and its (sub)multiples, before drawing these angles with compass and straight edge. In another activity we studied a fixed volume of water in various glass jars available in the laboratory. We studied the relationship between the height of water in these jars and the areas of cross section of the jars. Could the measure of volume in a conical jar have a formula? Since most classrooms in India are multilingual, we explored learning mathematics through language. What were the terms for buying, selling or the traditional names for money? In the grades where I taught both mathematics and science, it was possible to combine concepts e.g. a weighing scale as an equation or a suitable scale to measure and model the solar system. Could the same scale suffice for taking into account planet size on one hand as well as planetary distances on another?

In a normative practice of questioning and providing challenge, we as teachers learnt from each other too. On one occasion I had asked my colleagues:

Find the difference between 825 and its reverse. Is it a multiple of 9?

Find the difference between the two-digit number 'ab' and its reverse. How can you say whether it is or not a multiple of 9?

Now express the difference between the three-digit number 'abc' and its reverse.

Can we find a formula for the difference between any number and its reverse?

Not ceasing to inquire was an infectious paradigm to be a part of, though at times our attempts and experiments had no takers. Administration and routine have their place in schooling and took precedence on occasion.

Co-challenged

By way of summary I argue that the environment in our school enabled the teacher, the student and the subject of mathematics to be co-challenged. In our practice of morning assembly we implicitly conveyed what we gathered together in school for; to discuss as a collective with individual knowing. It is over time that the morning assembly problems addressed teaching-learning difficulties, set the day to a 'thinking' start and invited challenge among students. In my practice of classroom teaching-learning it was possible to challenge the banal in evaluation and invite students to be a part of mathematising by posing problems and not remain spectators or consumers of mathematics. In the kind of teaching that was mentored the premise of the need to experiment and arguing for ones attempts percolated to the students. Discovering concrete relationships within mathematics or with language and science was literally the last mile run in an institutional praxis embedded with questioning and debate.

The reactions and responses of our students spoke for our approach. While there existed ownership on one hand, there also existed voices of dissent in the culture that was constituted, lending opportunity for further debate. Such discussion also involved the parents of our students who were an intrinsic part of our school. One parent expressed the enjoyment she shared at calculating an interest of 150% over a $\frac{1}{4}$ day on a principal of Rs 1. Another wanted reason for examining his son on building a cube and not a paper-pencil test. Continued dialogue with them was as much a part of establishing and sustaining our school praxis.

Seizing the opportunity at a personal level led to my becoming a problem seeker and solver. My initiatives resulted in offshoots on the bus ride home or related problems on the pin-up board. In carrying forward my enthusiasm, I wrote short articles in a teacher's magazine and later compiled my efforts. In my writing (Gade, 2004) I argued for the learning of students to be dependent on the experience and opportunity made available to the students. It stemmed from my teaching practice in a school praxis which explicitly supported the value of recognising, accepting and addressing challenges.

A Vygotskian perspective

I conclude by reflecting on my experience and the environment in my school from a Vygotskian perspective. Vygotsky (1994) argued against the notion that the social environment be regarded as a condition of development and urged instead for insight of the relationships that existed between the child and the environment at a given stage of development. He maintained that the child's environment and the child both changed in development and that any social environment as a source of development possessed the ideal. With reference to the praxis in our school, the ideal of inquiry and never ceasing to question pervaded my practice of morning assemblies, classroom teaching-learning and teaching of mathematics across possible boundaries and barriers.

Baral and Das (2004) find Vygotsky's oeuvre of thinking compatible with that of Krishnamurti. They explain that for Krishnamurti intelligence is the harmony of reason, emotion and action and the act of becoming aware in one's relationship to the environment. This view they say echo's Vygotsky's view of the social-historical determinants of thought, language and action. Such a consensual view enables an analysis of the social environment I have outlined from two perspectives. Firstly, that social relationships bring about development and consciousness and secondly, that they also bring about self-awareness.

When applied to schooling the above arguments lead me to consider two other views. Firstly, Ivic (1994) suggests that the Vygotskian emphasis of education as an independent source of development needs to be explored and applied to schooling. Ivic allows me to argue that in the way we conceptualised our school and realised our ideals in multiple practices, our school was both medium and message and thereby an independent source of development. Secondly, Lemke (2002) argues that any culture not only shapes and prescribes what we need to learn, but also what ways of learning will be available to us. He argues that schooling is not only about what it is we study, but under the guidance of teachers, also about who it is we become. Lemke allows me to argue that in possessing and realising the ideal of questioning, our school implicitly urged our students to inquire and become independent thinkers.

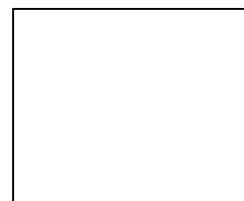
By drawing on the above I argue that the ideal of questioning and challenge in our school led to practices which allowed for the establishment of social relationships and their constant renegotiation. They also allowed the teacher to question routine and bring focus to the teaching-learning embedded within. Their opportunity to experiment in turn provided avenues for discovery of relationships within mathematics. Enabling change and continuous learning questioned the students' status quo resulting in their being challenged.

As I bring my arguments to a close I am conscious that I have not addressed the challenging nature of mathematics but the nature of schooling that can provide challenge in mathematics. I do this to bridge the divide and forward how the former can be brought about by the latter. I consider this issue worthy of debate since compulsory school is the last institution for many students anywhere in the world, where mathematics is learnt as a discipline and where it may be possible to not only challenge students but also their conception of mathematics.

A problem never dries up

In being a problem solver and poser one belongs to a nomadic tribe: taking problems to other places and collecting problems from there. Preparing to conduct a year long study, I shared a few problems with the two teachers whose class I was to observe. The teachers laid special focus on group work and collaborative learning by students and employed the following two problems at the beginning of the year. The first which I titled 'When together' asked:

In the pentagon are two dots, black and white, on the move. The black moves two corners counter clockwise. The white moves three corners clockwise. After how many moves are the two dots together?



The other problem titled ‘How heavy?’ was:

If a brick balances with three-quarters of a brick and three quarters of a pound, then how much does the brick weigh?

Someone challenged me with ‘When together’ and I do not know the source. I adapted ‘How heavy’ from a classic problem in Johnson. Herr and Kysh (2004) who in turn adapted the problem from ‘The weight of a brick’ in Loyd (1960).

During fieldwork my teachers announced and encouraged their students to participate in the Nordic competition in mathematics for school classes, the KappAbel (<http://www.kappabel.com/index.html>). In preparing for the competition, two students decided to check the researcher in their midst out and shared with me two problems from the University of Toronto Mathematics Network (<http://www.math.toronto.edu/mathnet/talent>). I shared the first (Problem 5/20) with my colleagues in the doctoral course on problem solving that I was parallelly attending. The students were able to solve the second (Problem 4/16), when it was redrawn and cut out on paper. These were not the only problems we challenged each other with. Another problem I set was:

Find a ten digit number in which the digit in any place tells you the number of such digits in the number, where the ten places in the digit from left to right correspond to the ten digits from 0 to 9 that can be used in the number.

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